

MENGZI, China

WMO: 569850

Lat: 23.4444N

Lon: 103.3278E

Elev: 1314

StdP: 86.51

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.6	6.1	-1.5	3.9	17.5	0.5	4.6	15.4	9.3	17.3	8.1	17.3	1.7	180	0.293

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	6.3	31.0	18.3	30.0	18.9	29.1	19.1	22.0	27.0	21.6	26.4	21.3	25.9	3.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.7	18.1	23.5	20.3	17.7	23.3	20.0	17.3	23.2	71.7	27.0	70.3	26.6	69.0	26.1	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.4	5.5	2.1	32.9	2.0	1.0	0.7	33.6	-0.4	34.2	-1.5	34.8	-3.0	35.5		
			1.1	22.6	2.0	0.6	-0.3	23.0	-1.4	23.3	-2.5	23.7	-3.9	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.0	13.6	15.9	19.3	22.2	23.5	24.3	23.9	23.6	22.6	20.4	17.1	13.9
	DBStd	4.71	3.30	4.14	3.62	3.08	2.79	1.88	1.50	1.54	1.97	2.36	2.67	2.92
	HDD10.0	24	10	7	1	0	0	0	0	0	0	1	6	
	HDD18.3	476	148	82	33	7	3	0	0	2	11	51	139	
	CDD10.0	3692	122	173	289	367	418	428	431	422	379	322	214	126
	CDD18.3	1102	1	16	63	124	162	179	172	164	131	74	15	1
	CDH23.3	5716	9	94	396	832	1052	1009	789	732	532	219	47	4
	CDH26.7	1183	1	6	72	243	304	212	134	124	72	12	1	0
Wind	WSAvg	2.5	2.8	3.1	3.2	3.2	2.8	2.5	2.1	1.8	2.0	2.3	2.2	2.4
Precipitation	PrecAvg	844	16	15	24	51	90	125	164	154	89	56	41	17
	PrecMax	1193	65	70	90	143	208	335	321	323	195	238	162	120
	PrecMin	511	0	0	0	9	22	36	42	18	23	0	0	0
	PrecStd	154	16	17	22	31	47	58	64	62	37	46	39	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.6	27.6	30.1	31.9	32.6	31.2	30.6	30.5	29.7	28.0	26.3	23.8
		MCWB	12.1	13.1	14.1	16.1	18.0	20.7	21.0	21.0	20.4	19.3	16.4	14.1
	2%	DB	22.7	25.7	28.6	30.7	31.1	30.0	29.3	29.2	28.5	26.7	24.7	22.2
		MCWB	12.4	12.9	14.3	16.5	18.5	20.7	20.9	21.0	20.2	18.7	16.0	13.9
	5%	DB	21.2	24.5	27.3	29.5	29.8	28.9	28.2	28.1	27.4	25.7	23.3	20.6
		MCWB	12.3	13.0	14.3	16.8	18.8	20.7	20.9	20.8	19.9	18.3	15.6	13.6
	10%	DB	19.6	22.9	25.7	28.0	28.4	27.9	27.1	27.0	26.4	24.6	21.9	19.0
		MCWB	12.2	12.9	14.5	16.9	19.0	20.5	20.8	20.7	19.6	18.0	15.2	13.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.9	16.2	17.8	20.1	21.4	22.2	22.5	22.6	21.8	20.4	18.5	16.2
		MCDB	19.5	22.5	23.9	26.4	27.9	27.9	27.6	27.6	26.6	25.6	23.3	21.3
	2%	WB	14.0	15.3	17.1	19.3	20.8	21.7	22.0	22.0	21.2	19.7	17.6	15.1
		MCDB	19.2	21.0	22.9	25.4	27.0	27.2	26.7	26.6	26.0	24.6	22.0	19.8
	5%	WB	13.3	14.6	16.5	18.8	20.4	21.3	21.6	21.5	20.8	19.2	16.8	14.2
		MCDB	18.8	20.4	22.3	24.8	26.2	26.6	26.0	26.0	25.3	23.7	21.0	19.1
	10%	WB	12.7	14.0	15.9	18.2	20.0	21.0	21.2	21.1	20.3	18.7	16.1	13.4
		MCDB	18.1	20.1	21.8	24.4	25.5	26.0	25.4	25.3	24.6	22.9	20.2	18.1
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.2	9.4	8.9	7.6	6.3	6.1	6.6	6.9	6.7	8.1	8.1
		MCDBR	11.9	11.9	11.8	10.6	9.3	7.8	7.8	8.3	8.4	8.5	10.1	10.8
		MCWBR	4.2	3.7	3.4	2.8	2.3	2.0	2.3	2.4	2.6	2.7	3.5	4.2
	5% WB	MCDBR	9.8	9.6	9.4	8.9	7.8	6.7	6.6	7.2	7.1	7.1	7.9	9.1
		MCWBR	4.1	3.5	3.4	3.0	2.4	2.1	2.4	2.5	2.5	2.5	3.2	3.9
Clear-Sky Solar Irradiance	taub	0.406	0.437	0.627	0.590	0.447	0.426	0.430	0.472	0.468	0.482	0.422	0.434	
	taud	2.167	2.089	1.620	1.770	2.253	2.346	2.316	2.173	2.160	2.062	2.171	2.062	
	Ebn at Noon	844	843	703	743	853	864	861	827	825	790	821	793	
	Edh at Noon	135	155	261	228	140	127	131	151	150	158	133	145	
All-Sky Solar Radiation	RadAvg	3.46	4.52	4.91	5.49	5.25	4.46	4.35	4.34	4.12	3.40	3.67	3.11	
	RadStd	0.44	0.68	0.42	0.51	0.59	0.40	0.27	0.22	0.25	0.38	0.30	0.36	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.51	N/A	N/A	+0.62	N/A	N/A	N/A	N/A	+193	+147
(47)	Regional (6 neighbors)	+0.37	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	+106	+60

Nomenclature: See separate page